

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12417661
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Creque; Andrew J. et al.

Theft deterring product display

Abstract

A secure product display system and associated method can include a display secured with a lock, a dispenser including a digital key request code and an input configured to accept entry of a digital key, and an access cloud configured to transmit the digital key to a user device when the digital key is requested using the digital key request code, wherein the lock is unlocked when the digital key, entered with the input, is authentic.

Inventors: Creque; Andrew J. (Macedonia, OH), Lanese; Gina M. (Stow, OH), Myler; Jeffry A. (Stow, OH), Forchione; Allan N. (Canton, OH), Kollar; Troy (Parma, OH)

Applicant: Creque; Andrew J. (Macedonia, OH); Lanese; Gina M. (Stow, OH); Myler; Jeffry A. (Stow, OH); Forchione; Allan N. (Canton, OH); Kollar; Troy (Parma, OH)

Family ID: 1000008822196

Assignee: DARKO, INC. (Bedford Heights, OH)

Appl. No.: 18/199684

Filed: May 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230377392 A1	Nov. 23, 2023

Related U.S. Application Data

us-provisional-application US 63344515 20220520

Publication Classification

Int. Cl.: G07C9/00 (20200101); G06F21/36 (20130101)

U.S. Cl.:

CPC **G07C9/00182** (20130101); **G06F21/36** (20130101); **G07C9/0069** (20130101);
G07C2009/00253 (20130101); G07C2009/00277 (20130101)

Field of Classification Search

CPC: G07C (9/00182); G07C (9/0069); G07C (2009/00253); G07C (2009/00277); G06F
(21/36)

USPC: 340/5.6

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5194856	12/1992	Zijlstra	N/A	N/A
5600121	12/1996	Kahn et al.	N/A	N/A
5635906	12/1996	Joseph	N/A	N/A
5713485	12/1997	Liff et al.	N/A	N/A
5745036	12/1997	Clare	N/A	N/A
5887176	12/1998	Griffith et al.	N/A	N/A
6169483	12/2000	Ghaffari et al.	N/A	N/A
6272394	12/2000	Lipps	N/A	N/A
6338007	12/2001	Broadfield et al.	N/A	N/A
6681989	12/2003	Bodin	N/A	N/A
7150365	12/2005	Hardy et al.	N/A	N/A
7451881	12/2007	Hardy et al.	N/A	N/A
7497341	12/2008	Hardy et al.	N/A	N/A
7513419	12/2008	Crews	235/379	G06Q 20/042
7635068	12/2008	Hardy	N/A	N/A
8047385	12/2010	Hardy	N/A	N/A
8120468	12/2011	Kangas	N/A	N/A
8210363	12/2011	Hardy	N/A	N/A
8235227	12/2011	Hardy	N/A	N/A
8413823	12/2012	Hardy	N/A	N/A
8727179	12/2013	Colelli et al.	N/A	N/A
8787902	12/2013	Kim	455/426.1	H04W 12/068
9277831	12/2015	Hardy	N/A	N/A
9526351	12/2015	Hardy	N/A	N/A
9552710	12/2016	Rasband et al.	N/A	N/A
9659424	12/2016	Huber et al.	N/A	N/A
9687085	12/2016	Hardy	N/A	N/A
9706857	12/2016	Hardy et al.	N/A	N/A
9723934	12/2016	Hardy	N/A	N/A
9844280	12/2016	Dipalo et al.	N/A	N/A
9993091	12/2017	Hardy	N/A	N/A
10026248	12/2017	Woodard et al.	N/A	N/A
10051977	12/2017	Hardy	N/A	N/A

10062257	12/2017	Chau	N/A	N/A
10332117	12/2018	Krause et al.	N/A	N/A
10349755	12/2018	Dipaolo et al.	N/A	N/A
10441092	12/2018	Hardy	N/A	N/A
10555624	12/2019	Hardy	N/A	N/A
10667630	12/2019	Hardy	N/A	N/A
10905258	12/2020	Hardy et al.	N/A	N/A
10945538	12/2020	Hardy	N/A	N/A
11051637	12/2020	Hardy	N/A	N/A
11058234	12/2020	Dipaolo et al.	N/A	N/A
11109692	12/2020	Swafford	N/A	N/A
11178982	12/2020	Hardy	N/A	N/A
11182803	12/2020	Budano	N/A	G06K 7/1417
11188973	12/2020	Bynum et al.	N/A	N/A
11216827	12/2021	Budano	N/A	N/A
11344138	12/2021	Hardy et al.	N/A	N/A
11375826	12/2021	Hardy	N/A	N/A
11587081	12/2022	Weston	N/A	G06Q 20/382
11599891	12/2022	Budano	N/A	N/A
11631089	12/2022	Budano	N/A	N/A
11887131	12/2023	Budano et al.	N/A	N/A
11928694	12/2023	Budano et al.	N/A	N/A
2002/0016715	12/2001	Razumov	N/A	N/A
2004/0079799	12/2003	Symonds	235/381	G06Q 30/06
2006/0022827	12/2005	Higham	N/A	N/A
2008/0001747	12/2007	Kangas	N/A	N/A
2009/0322529	12/2008	Kangas	N/A	N/A
2010/0187307	12/2009	Phillips et al.	N/A	N/A
2011/0074543	12/2010	Kaczmarz et al.	N/A	N/A
2011/0133948	12/2010	Ervin	340/687	A61J 7/0409
2015/0143123	12/2014	Bergqvist	713/171	B67D 7/348
2015/0379366	12/2014	Nomura et al.	N/A	N/A
2016/0260155	12/2015	Landsman	N/A	G06Q 30/0631
2017/0217011	12/2016	Savage et al.	N/A	N/A
2017/0372555	12/2016	Clark et al.	N/A	N/A
2018/0374327	12/2017	Enekwa et al.	N/A	N/A
2021/0395070	12/2020	Sankaran	N/A	G05B 19/4155
2022/0041425	12/2021	Sankaran	N/A	G06F 3/0482
2023/0129525	12/2022	Lee	426/231	A47J 31/521
2023/0153831	12/2022	Budano et al.	N/A	N/A
2024/0042888	12/2023	Goswami	N/A	H02J 7/0048
2024/0104582	12/2023	Budano et al.	N/A	N/A

Primary Examiner: Casillashernandez; Omar

Attorney, Agent or Firm: LOZA & LOZA, LLP

Background/Summary

CROSS REFERENCE TO RELATED PATENT APPLICATIONS (1) The present application claims the priority and benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/344,515 filed May 20, 2022, entitled "THEFT DETERRING PRODUCT DISPLAY." U.S. Provisional Patent Application Ser. No. 63/344,515 is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(1) Embodiments are generally related to the field of product displays. Embodiments are related to locking devices. Embodiments are also related to the field of theft deterrence. Embodiments are further related to the field of computer devices and mobile devices used for providing digital keys to open secure product displays. Embodiments are also related to methods, systems, and devices for deterring retail theft with locking displays which can be unlocked with digital keys.

BACKGROUND

(2) A critical aspect of retail sales is the display of merchandise in a manner that allows customers to peruse offerings. In many cases, this involves the time tested practice of holding merchandise on pegs, shelves, or display stands openly accessible to customers. Open access to merchandise allows customers to review products at their leisure, but also comes at a cost, as it leaves the items vulnerable to theft.

(3) Profit loss due to theft is nothing new in retail. In response, retailers use a wide variety of theft deterrent and prevention solutions. Some theft deterrent solutions make the process of stealing so uncomfortable (e.g., slow) that thieves are unwilling to take the risk. A common theft prevention solution involve placing certain items under lock-and-key, making the process of stealing sufficiently inconvenient and conspicuous, that it is not worth the risk.

(4) Small merchandise is particularly attractive to a thief because it is easy to conceal. The value of a small retail item determines which type of security solution makes sense. For example, jewelry is both small and expensive and is therefore usually stored in a theft prevention device such as a locked display cabinet or jewelry safe as illustrated in FIGS. 1A and 1B labeled as prior art.

(5) Other less expensive items, such razors, or batteries, that are still valuable enough to be attractive to a thief are usually secured using a theft deterrent device such as an anti-sweep peg board spiral as shown in FIG. 2 labeled as prior art.

(6) Obviously, it would be possible to reduce theft, regardless of item type, by securing everything in a theft prevention device as illustrated in FIGS. 1 and 2. Unfortunately, placing every item under lock-and-key is expensive. In addition, shoppers don't appreciate the inconvenience associated with finding a store associate to unlock items for them. This leads to lost sales and can discourage the shoppers so much that they elect to shop elsewhere, where making selections does not require a store associate to disarm an anti-theft device.

(7) Prior art solutions, as described above, have various short comings. Locking anything using a physical key and lock is problematic for retailers. Keys frequently get lost or copied and locks get damaged. These problems individually are of little consequence, but collectively demand valuable retailer resources to resolve and maintain. Similarly anti-sweep devices slow down theft, but when theft occurs there is no way to identify the thief.

(8) Furthermore, some prior art solutions include digitally controlled distribution systems. These systems are convenient when they are operational. However, most such systems are reliant on internet of things (IOT) devices, which require an internet connection. It can be expensive and time consuming to maintain internet configurations associated with retail displays. When the retail internet or IOT device is inoperative or unavailable, the digitally controlled distribution system is similarly inoperable.

(9) Accordingly, there is a need in the art for methods and systems that serve to deter theft without deterring legitimate customers from making purchases as detailed in the embodiments herein.

SUMMARY

(10) The following summary is provided to facilitate an understanding of some of the innovative features unique to the embodiments disclosed and is not intended to be a full description. A full appreciation of the various aspects of the embodiments can be gained by taking the entire specification, claims, drawings, and abstract as a whole.

(11) It is, therefore, one aspect of the disclosed embodiments to provide improved methods and systems for theft deterrence.

(12) It is another aspect of the disclosed embodiments to provide a method, system, and apparatus for retail theft deterrence.

(13) It is another aspect of the disclosed embodiments to provide methods, systems, and apparatuses for product displays with integrated theft deterrence mechanisms.

(14) It is another aspect of the disclosed embodiments to provide methods, systems, and apparatuses for product displays with integrated theft deterrence mechanisms, which can be opened with digital keys.

(15) In the embodiments herein, a system, method, and apparatus for a theft deterring retail display system including a retail display fitted with an electronic lock. The electronic lock can be unlocked with a digital key. The digital key can be dispensed electronically to a customer when the customer provides customer data. The system can include a validity check to ensure the data provide by the customer is valid and authentic before dispensing the digital key.

(16) For example, in an embodiment, a display system comprises a display secured with a lock, a dispenser, the dispenser comprising: a digital key request code and an input configured to accept entry of a digital key, and an access cloud configured to transmit the digital key to a user device when the digital key is requested using the digital key request code, wherein the lock is unlocked when the digital key, entered with the input, is authentic. In an embodiment, the lock comprises an electronic lock. In an embodiment, the digital key request code comprises a QR code. In an embodiment, the digital key comprises a string of numbers. In an embodiment, the digital key comprises a unique digital key every time the digital key is requested using the digital key request code. In an embodiment, the display comprises a retail product display. In an embodiment, the access cloud further comprises a mobile UI configured to require the user to input their phone number in order to receive the digital key. In an embodiment, the mobile UI provides the digital key to the user after the user enters their phone number. In an embodiment, the access cloud provides the digital key to the user as a text message using the phone number provided by the user.

(17) In another embodiment, a secure display system comprises a display secured with a lock, a digital key request code, an access module configured to transmit a digital key to a user device when a digital key is requested using the digital key request code, and a digital key reader configured to unlock the lock when the digital key is authentic. In an embodiment, the lock comprises an electronic lock. In an embodiment, the display comprises a retail product display. In an embodiment, the digital key request code comprises a QR code. In an embodiment, the digital key request code is provided proximate to the digital key reader. In an embodiment, the access module generates a unique digital key every time the digital key is requested using the digital key request code. In an embodiment, the digital key comprises a QR code. In an embodiment, the digital key reader comprises a camera configured to scan the QR code.

(18) In another embodiment, a secure access method comprises providing a digital key request code on a dispenser, upon receiving a digital key request, requiring entry of a phone number via a web portal, providing a digital key to a user device, accepting input at the dispenser, the input comprising the digital key, authenticating the digital key, and unlocking a lock associated with a display when the digital key is authentic. In an embodiment, providing a digital key to a user device further comprises at least one of: providing the digital key to the user via a mobile UI and/or providing the digital key to the user as a text message using the phone number entered via the web portal. In an embodiment, the digital key comprises at least one of a QR code and/or a string of numbers.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The accompanying figures, in which like reference numerals refer to identical or functionally similar elements throughout the separate views and which are incorporated in and form a part of the specification, further illustrate the embodiments and, together with the detailed description, serve to explain the embodiments disclosed herein.
- (2) FIG. 1A illustrates prior art theft deterrence means;
- (3) FIG. 1B illustrates prior art theft deterrence means;
- (4) FIG. 2 illustrates prior art theft deterrence means;
- (5) FIG. 3 depicts a block diagram of a computer system which is implemented in accordance with the disclosed embodiments;
- (6) FIG. 4A depicts a graphical representation of a network of data-processing devices in which aspects of the present embodiments may be implemented;
- (7) FIG. 4B depicts a graphical representation of a network associated with a digital key dispenser in which aspects of the present embodiments may be implemented;
- (8) FIG. 5 depicts a computer software system for directing the operation of the data-processing system depicted in FIG. 1, in accordance with an example embodiment;
- (9) FIG. 6 depicts a block diagram of a display security system, in accordance with the disclosed embodiments;
- (10) FIG. 7 depicts an access module, in accordance with the disclosed embodiments;
- (11) FIG. 8A depicts a digital key dispenser, in accordance with disclosed embodiments;
- (12) FIG. 8B depicts a home screen of a user interface of a user portal displayed on a user device, in accordance with the disclosed embodiments;
- (13) FIG. 8C depicts a block diagram of a digital key dispensation method, in accordance with the disclosed embodiments;
- (14) FIG. 8D depicts a retail display with an associated digital key dispenser, in accordance with the disclosed embodiments;
- (15) FIG. 9 depicts a block diagram of a store module, in accordance with the disclosed embodiments;
- (16) FIG. 10 depicts a block diagram of another display security system, in accordance with the disclosed embodiments;
- (17) FIG. 11 depicts a block diagram of a gateway, in accordance with the disclosed embodiments;
- (18) FIG. 12 depicts a block diagram of a smart lock, in accordance with the disclosed embodiments;
- (19) FIG. 13 depicts steps associated with a method for deterring theft, in accordance with the disclosed embodiments;
- (20) FIG. 14 depicts another product access system, in accordance with the disclosed embodiments;
- (21) FIG. 15 depicts an exemplary screen shot of a web portal home page associated with a product access system, in accordance with the disclosed embodiments;
- (22) FIG. 16 depicts aspects of a dispenser associated with a product access system, in accordance with the disclosed embodiments; and
- (23) FIG. 17 depicts an exemplary lock associated with a product access system, in accordance with the disclosed embodiments.

DETAILED DESCRIPTION

- (24) The particular values and configurations discussed in the following non-limiting examples can be varied, and are cited merely to illustrate one or more embodiments and are not intended to limit the scope thereof.
- (25) Example embodiments will now be described more fully hereinafter, with reference to the

accompanying drawings, in which illustrative embodiments are shown. The embodiments disclosed herein can be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the embodiments to those skilled in the art. Like numbers refer to like elements throughout.

(26) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(27) Throughout the specification and claims, terms may have nuanced meanings suggested or implied in context beyond an explicitly stated meaning. Likewise, the phrase “in one embodiment” as used herein does not necessarily refer to the same embodiment and the phrase “in another embodiment” as used herein does not necessarily refer to a different embodiment. It is intended, for example, that claimed subject matter include combinations of example embodiments in whole or in part.

(28) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(29) It is contemplated that any embodiment discussed in this specification can be implemented with respect to any method, kit, reagent, or composition of the invention, and vice versa.

Furthermore, compositions of the invention can be used to achieve methods of the invention.

(30) It will be understood that particular embodiments described herein are shown by way of illustration and not as limitations of the invention. The principal features of this invention can be employed in various embodiments without departing from the scope of the invention. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, numerous equivalents to the specific procedures described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.

(31) The use of the word “a” or “an” when used in conjunction with the term “comprising” in the claims and/or the specification may mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.” The use of the term “or” in the claims is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.” Throughout this application, the term “about” is used to indicate that a value includes the inherent variation of error for the device, the method being employed to determine the value, or the variation that exists among the study subjects.

(32) As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or method steps.

(33) The term “or combinations thereof” as used herein refers to all permutations and combinations of the listed items preceding the term. For example, “A, B, C, or combinations thereof” is intended to include at least one of: A, B, C, AB, AC, BC, or ABC, and if order is important in a particular context, also BA, CA, CB, CBA, BCA, ACB, BAC, or CAB. Continuing with this example, expressly included are combinations that contain repeats of one or more item or term, such as BB,

AAA, AB, BBC, AAABCCCC, CBBAAA, CABABB, and so forth. The skilled artisan will understand that typically there is no limit on the number of items or terms in any combination, unless otherwise apparent from the context.

(34) All of the compositions and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of preferred embodiments, it will be apparent to those of skill in the art that variations may be applied to the compositions and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit, and scope of the invention. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the invention as defined by the appended claims.

(35) FIGS. 3-5 are provided as exemplary diagrams of data-processing environments in which embodiments of the present invention may be implemented. It should be appreciated that FIGS. 3-5 are only exemplary and are not intended to assert or imply any limitation with regard to the environments in which aspects or embodiments of the disclosed embodiments may be implemented. Many modifications to the depicted environments may be made without departing from the spirit and scope of the disclosed embodiments.

(36) A block diagram of a computer system **300** that executes programming for implementing parts of the methods and systems disclosed herein is shown in FIG. 3. A computing device in the form of a computer **310** configured to interface with sensors, peripheral devices, and other elements disclosed herein may include one or more processing units **302**, memory **304**, removable storage **312**, and non-removable storage **314**. Memory **304** may include volatile memory **306** and non-volatile memory **308**. Computer **310** may include or have access to a computing environment that includes a variety of transitory and non-transitory computer-readable media such as volatile memory **306** and non-volatile memory **308**, removable storage **312** and non-removable storage **314**. Computer storage includes, for example, random access memory (RAM), read only memory (ROM), erasable programmable read-only memory (EPROM) and electrically erasable programmable read-only memory (EEPROM), flash memory or other memory technologies, compact disc read-only memory (CD ROM), Digital Versatile Disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage, or other magnetic storage devices, or any other medium capable of storing computer-readable instructions as well as data including image data.

(37) Computer **310** may include or have access to a computing environment that includes input **316**, output **318**, and a communication connection **320**. The computer may operate in a networked environment using a communication connection **320** to connect to one or more remote computers, remote sensors, actuators, smart locks, detection devices, hand-held devices, multi-function devices (MFDs), mobile devices, tablet devices, mobile phones, Smartphones, or other such devices. The remote computer may also include a personal computer (PC), server, router, network PC, RFID enabled device, a peer device or other common network node, or the like. The communication connection may include a Local Area Network (LAN), a Wide Area Network (WAN), Bluetooth connection, or other networks. This functionality is described more fully in the description associated with FIG. 4 below.

(38) Output **318** is most commonly provided as a computer monitor, but may include any output device. Output **318** and/or input **316** may include a data collection apparatus associated with computer system **300**. In addition, input **316**, which commonly includes a computer keyboard and/or pointing device such as a computer mouse, computer track pad, or the like, allows a user to select and instruct computer system **300**. A user interface can be provided using output **318** and input **316**. Output **318** may function as a display for displaying data and information for a user, and for interactively displaying a graphical user interface (GUI) **330**.

(39) Note that the term “GUI” generally refers to a type of environment that represents programs,

files, options, and so forth by means of graphically displayed icons, menus, and dialog boxes on a computer monitor screen. A user can interact with the GUI to select and activate such options by directly touching the screen and/or pointing and clicking with a user input device **316** such as, for example, a pointing device such as a mouse and/or with a keyboard. A particular item can function in the same manner to the user in all applications because the GUI provides standard software routines (e.g., module **325**) to handle these elements and report the user's actions. The GUI can further be used to display the electronic service image frames as discussed below.

(40) Computer-readable instructions, for example, program module or node **325**, which can be representative of other modules or nodes described herein, are stored on a computer-readable medium and are executable by the processing unit **302** of computer **310**. Program module or node **325** may include a computer application. A hard drive, CD-ROM, RAM, Flash Memory, and a USB drive are just some examples of articles including a computer-readable medium.

(41) FIG. **4A** depicts a graphical representation of a network of data-processing systems **400** in which aspects of the present invention may be implemented. Network data-processing system **400** is a network of computers or other such devices such as mobile phones, smartphones, sensors, detection devices, and the like in which embodiments of the present invention may be implemented. Note that the system **400** can be implemented in the context of a software module such as program module **325**. The system **400** includes a network **402** in communication with one or more clients **410**, **412**, and **414**, and lock **405**. Network **402** may also be in communication with one or more external devices such as digital key dispensers **404**, sensors, or locks **405**, servers **406**, and storage **408**. Network **402** is a medium that can be used to provide communications links between various devices and computers connected together within a networked data processing system such as computer system **300**. Network **402** may include connections such as wired communication links, wireless communication links of various types, fiber optic cables, quantum, or quantum encryption, or quantum teleportation networks, etc. It should be understood that external device **404** may be embodied as a lock, electronic lock, mobile device, cell phone, tablet device, monitoring device, detector device, sensor, microcontroller, controller, receiver, transceiver, or other such device.

(42) In the depicted example, external device **404**, server **406**, and clients **410**, **412**, and **414** connect to network **402** along with storage unit **408**. Clients **410**, **412**, and **414** may be, for example, personal computers or network computers, handheld devices, mobile devices, tablet devices, smartphones, personal digital assistants, microcontrollers, recording devices, MFDs, etc. Computer system **300** depicted in FIG. **3** can be, for example, a client such as client **410** and/or **412**.

(43) Computer system **300** can also be implemented as a server such as server **406**, depending upon design considerations. In the depicted example, server **406** provides data such as boot files, operating system images, applications, and application updates to clients **410**, **412**, and/or **414**. Clients **410**, **412**, and **414** and external device **404** are clients to server **406** in this example. Network data-processing system **400** may include additional servers, clients, and other devices not shown. Specifically, clients may connect to any member of a network of servers, which provide equivalent content.

(44) In the depicted example, network data-processing system **400** is the Internet with network **402** representing a worldwide collection of networks and gateways that use the Transmission Control Protocol/Internet Protocol (TCP/IP) suite of protocols to communicate with one another. At the heart of the Internet is a backbone of high-speed data communication lines between major nodes or host computers consisting of thousands of commercial, government, educational, and other computer systems that route data and messages. Of course, network data-processing system **400** may also be implemented as a number of different types of networks such as, for example, an intranet, a local area network (LAN), or a wide area network (WAN). FIGS. **3** and **4** are intended as examples and not as architectural limitations for different embodiments of the present invention.

(45) FIG. 5 illustrates a software system **500**, which may be employed for directing the operation of the data-processing systems such as computer system **300** depicted in FIG. 3. Software application **505**, may be stored in memory **304**, on removable storage **312**, or on non-removable storage **314** shown in FIG. 3, and generally includes and/or is associated with a kernel or operating system **510** and a shell or interface **515**. One or more application programs, such as module(s) or node(s) **325**, may be “loaded” (i.e., transferred from removable storage **314** into the memory **304**) for execution by the data-processing system **500**. The data-processing system **500** can receive user commands and data through user interface **515**, which can include input **316** and output **318**, accessible by a user **520**. These inputs may then be acted upon by the computer system **500** in accordance with instructions from operating system **510** and/or software application **505** and any software module(s) **325** thereof.

(46) Generally, program modules (e.g., module **325**) can include, but are not limited to, routines, subroutines, software applications, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types and instructions. Moreover, those skilled in the art will appreciate that elements of the disclosed methods and systems may be practiced with other computer system configurations such as, for example, hand-held devices, mobile phones, smart phones, tablet devices, multi-processor systems, printers, copiers, fax machines, multi-function devices, data networks, microprocessor-based or programmable consumer electronics, networked personal computers, minicomputers, mainframe computers, servers, medical equipment, medical devices, and the like.

(47) Note that the term module or node as utilized herein may refer to a collection of routines and data structures that perform a particular task or implements a particular abstract data type. Modules may be composed of two parts: an interface, which lists the constants, data types, variables, and routines that can be accessed by other modules or routines; and an implementation, which is typically private (accessible only to that module), and which includes source code that actually implements the routines in the module. The term module may also simply refer to an application such as a computer program designed to assist in the performance of a specific task such as word processing, accounting, inventory management, etc., or a hardware component designed to equivalently assist in the performance of a task.

(48) The interface **515** (e.g., a graphical user interface **330**) can serve to display results, whereupon a user **520** may supply additional inputs or terminate a particular session. In some embodiments, operating system **510** and GUI **330** can be implemented in the context of a “windows” system. It can be appreciated, of course, that other types of systems are possible. For example, rather than a traditional “windows” system, other operation systems such as, for example, a real time operating system (RTOS) more commonly employed in wireless systems may also be employed with respect to operating system **510** and interface **515**. The software application **505** can include, for example, module(s) **525**, which can include instructions for carrying out steps or logical operations such as those shown and described herein.

(49) The following description is presented with respect to embodiments of the present invention, which can be embodied in the context of, or require the use of a data-processing system such as computer system **300**, in conjunction with program module **325**, and data-processing system **400** and network **402** depicted in FIGS. 3-5. The present invention, however, is not limited to any particular application or any particular environment. Instead, those skilled in the art will find that the systems and methods of the present invention may be advantageously applied to a variety of system and application software including database management systems, word processors, and the like. Moreover, the present invention may be embodied on a variety of different platforms including Windows, Macintosh, UNIX, LINUX, Android, Arduino, and the like. Therefore, the descriptions of the exemplary embodiments, which follow, are for purposes of illustration and not considered a limitation.

(50) In the embodiments herein, a system, method, and apparatus comprise a display system

configured to deter theft. It should be appreciated that the display disclosed herein can comprise a retail display. However, in other embodiments, the display can comprise a display configured for other purposes. The embodiments include a blend of a theft deterrent and theft protection solutions. The embodiments keeps items locked up but allow shoppers access to the key without the direct assistance of a retail associate. In some cases, the embodiments are configured to operate without the use of a retail internet connection, requiring only that the customer have an internet connected device (e.g., a smart phone). It is advantageous to reduce the burden of maintaining internet connected retail devices.

(51) FIG. 6 illustrates a block diagram of a display system **600** for displaying products with integrated theft deterrence in accordance with the disclosed embodiments. The system can generally include a mobile device, or user device **605** that provides a user portal **610**. The user portal **610** can serve as the interface through which the customer interacts with the system **600**.

(52) As illustrated in FIG. 6, the user device **605** can be any device that is equipped with a camera **615** or other such interface where a code can be entered, and internet connection. In certain embodiments, the user device **605** can be configured to read QR codes or near field communication (NFC) and can have a GPS receiver **620**, such that the user device **605** can verify information from a QR code or NFC, and can verify position and time. In certain embodiments, the user device **605** can comprise a mobile device, or smart phone, a tablet device, a computer, a smart watch, a smart wearable device, or other such device.

(53) The user device **605** can be equipped with a user portal **610**. In certain cases, the user portal **610** can comprise a standalone software module (e.g., an app). In other embodiments, the user portal **610** can be a web portal served to a webpage operated on the user device **605**. The user portal **610** can serve as an interface through which the user interacts with the product display system **600**. The interface can comprise a specially designed graphical user interface that allows the user to receive alerts relating to the display system. It should be understood that in certain embodiments, the display system can interact with the user device via SMS messaging, via an internet browsing application provided on the device, or other such digital protocols only. This can be achieved in a number of ways as further detailed herein.

(54) The user device **605** can interact with other aspects of the system through a wired or wireless connection via the world wide web, a Bluetooth connection, a near field connection, or the like. In certain embodiments, this can be achieved using a cloud computing architecture. In certain embodiments, a server **625** can provide data **635** storage and can provide an access module **630**. Data storage **635** can include structured data storage for all the data collected herein, as further detailed below. The data storage can be accessed from the gateway **416**, store portal **650**, or both according to the associated system architecture for a given installation.

(55) The display system **600** further includes a security device **640**. The security device **640** can be an electronic locking mechanism, including but not limited to a smart lock or other such electronic lock. The security device **640** can be configured to lock a product display **645**. The display can serve as a product display embodied as a cabinet, enclosed shelf, locked peg, or other such product display. The security device **640** can interact with the system **600** to allow access to the product display **645** and products **665** therein.

(56) FIG. 12 depicts aspects of the security device **640** embodied as a smart lock. The smart lock can include a power source **1205** which can preferably be a battery, although a wired power connection is also possible. Likewise, the smart lock **640** can include a wireless control module **1210**, eliminating the need for control wiring between key dispenser and the lock. The smart lock can further be optionally configured with a light **1215**, loudspeaker **1220**, and/or a spring biased release **1225** so that when the lock disengages, it does so in a manner that makes it obvious the associated door is unlocked; for example, a light will flash, a sound will play, and/or the door will pop open). The smart lock **640** can further include a sensor **1230** so that when a shopper walks away, the smart lock **640** can identify their departure and can check that the door is in the “closed”

position. The smart lock **640** and actively latch and lock itself (i.e., it will automatically lock itself and not require a shopper to close and lock it properly). If the smart lock determines it is not properly latched and locked it can provide a visible, audible, and/or wireless notification for store associates.

(57) In certain embodiments, numerous security devices **640** and/or product displays **645** can be distributed throughout a retail environment. In certain aspects, numerous systems architectures are possible using the embodiments disclosed herein. For example, FIG. **4B** illustrates that in certain embodiments, a gateway **416** can be used to provide the access module **630** via a website or other such internet accessible node. In such embodiments, one digital key dispenser **404** can be associated with one or more product displays **645** and associated locks **405** (or smart locks **640**). Numerous digital key dispensers **404** can be distributed in the retail environment as illustrated. Of note, the customer is able to access the gateway **416** and digital key dispenser via their own internet connected user device.

(58) The display system **600** can further include a store portal or other such monitoring station **650**, which can comprise a computer system (as illustrated in FIGS. **3-5**) wherein a retail store manager or administrator can control aspects of the disclosed system **600**. An interface associated with store module **655**, can comprise a specially designed graphical user interface that allows the administrator to set and receive alerts relating to scheduling, monitor the display, send messages to a user, unlock security devices **640** for restocking or other purposes, and set other system parameters. It should be understood that the store portal **650** can comprise multiple devices accessible by multiple administrators in certain embodiments. In certain embodiments, the store portal **650** can comprise any device with a browser (or app) used by authorized personnel to administer the overall system. Tasks such as managing the blocked list, approved list, generating usage reports, configuration notifications, and monitoring the overall system can be provided and completed via the store portal **650**.

(59) The access module **630** can comprise the root application (or software) that is used to control the system. The server **625** can serve data to, and accept data from, the user device **605**, the smart lock **640**, and the store module **655**. In other embodiments, cloud storage **660** can be used to store some or all of the relevant data and/or software associated with the system **600**.

(60) FIG. **7** illustrates a system diagram of the access module **630**. The access module **630** can include a digital key module **705**, authentication module **710**, and lock release module **715** and can serve as an API for accessing data stored in database **625**.

(61) In an exemplary embodiment, the digital key module **705** can control the distribution of digital keys to customers. The system **600** can include a digital key dispenser **404** as illustrated in FIG. **8**. The digital key dispenser can be disposed on a product display **645**, or in another location proximate to the product display **645**. The digital key dispenser can include a digital key request code **805**. In certain embodiments, this can be embodied as a QR code but in other embodiments other such codes can be used. In certain embodiments, the digital key request code **805** can be uniquely associated with the product display **645** to which it is attached, since a retail sales environment may include multiple secure product displays **645** throughout the retail space. Note the digital key request code **805** need not be a digital rendering. In certain embodiments, the digital key request code **805** can be printed on or near the associated display.

(62) The digital key request code **805** can be scanned or otherwise entered at a user device **605**. The digital key request code **805** can initiate an instances of a web browser which can automatically open on the user device **605**. FIG. **8B** illustrates aspects of a user portal **610**, which can comprise a web application, or other such software. In certain embodiments, the user portal **610** can also be provided as a downloadable application (app), although the use of an app is not required. The user portal **610** can be used to first prompt **820** the user to identify the door which needs to be unlocked so that the user can access a retail item inside.

(63) In addition, the user portal **610** can require the user to provide an identity verification **825**

which can be a phone number and, in certain cases other biographical or identifying data. In certain embodiments, this can include an email address, physical address, photograph, information on a driver's license, etc. The identifying information **825** required can be selected by the administrator. In certain embodiments, the detail required in the identifying information **825** may be related to the level of security for the retail item in the product display **645**.

(64) However, in certain embodiments, the system **600** is meant to serve as a theft deterrent that does not significantly inconvenience the customer. As such, the customer phone number may be the only identifying information required. When the user provides the identifying data, it can be stored as data **635**. Once the data is entered, the digital key module **705** can generate a unique digital key **810** which can be displayed via the user portal **610**. This process is illustrated in FIG. **8C**.

(65) The access module **630** can likewise provide the digital key **810**, or other such authenticating data, matched to the unique digital key served to the user portal **610**, to the authentication module **710**.

(66) The digital key dispenser **404** can further include a digital key reader **815**. The digital key reader **815** can comprise a camera, or other such key input. For example, in other embodiment, the digital key reader **815** can comprise an input where a user can manually enter aspects of the digital key **810**. When the digital key **810** is embodied as a QR code in the user portal **610**, the QR code can be held in the view of the digital key reader **815** camera. The digital key reader **815** can provide the digital key **810** to the authentication module **710**. The digital key stored in data **635** can then be used to authenticate the digital key provided via the user module **610**.

(67) If the digital key **810** is authenticated, the lock release module can send a signal to the smart lock **640** in order to unlock the product display **645** as illustrated in FIG. **8D**. If the digital key is not authenticated, the display can remain locked, and the user can be prompted to request a valid digital key via a display or loudspeaker configured on the digital key dispenser **404**.

(68) It should be appreciated that the digital key request code and/or the digital key can take many forms. In the embodiment description above a QR code is used for both. However, other digital formats also apply. For example, using NFC technology a shopper could tap their mobile device on the Digital Key Request Code device and then tap their mobile device on the Digital Key Reader device once they receive a Digital Key.

(69) The system **600** is configured for more than providing digital keys to shoppers. Each store portal **650** can be managed by a store manager, or other trusted associate (this person, or people, can be collectively referred to as an administrator). The store module **655** can be configured to allow the administrator to login into an admin portal **905** associated with their unique account as illustrated in FIG. **9**. The admin portal **905** provides many capabilities.

(70) For example, the store module can include a dashboard **910** summarizing the health of all the digital key reader devices (e.g., fixtures) associated with the store module **655**.

(71) In certain embodiment, tunable artificial intelligence module **915** can automatically detects nefarious shopper behavior associated with a specific mobile device **605** and can automatically add the device to a blocked list **920**. Those on the blocked list **920** using device **605** and/or associated phone numbers can be denied new digital keys. The blocked list **920** can be edited using the admin portal **905**. Likewise, an approved list **940** can be provided. Those on the approved list can include store personnel or trustworthy customers who open the display often. Those on the approved list will not be flagged for frequent use, or unusual behavior associated with the display.

(72) The admin portal **905** can be used to access a wide variety of usage reports **925** (i.e., shopper analytics). These can include specific analytics **925** relating to total purchases from a given display, time of purchase, demographics associated with the purchaser, and other such data.

(73) In certain embodiments, a branding module **930** allows the admin to brand messages, such as text message sent to the shopper containing the digital key. Likewise, the branding module can be used to provide coupon codes or other purchasing incentives with the digital key message.

(74) In certain embodiments, a security module **935** associated with the store module can be used

to provide notifications based on various user defined criteria. For example, one aspect of the security module **935** is to automatically transmit a notification such as a text message for events including but not limited to: new blocked list devices, attempts to obtain a digital key using a blocked listed device, a door left open for too long, removal of a large quantity of products with one digital key, and the like.

(75) Aspects of another embodiment of a retail display security system **1000** are illustrated in FIG. **10**. It should be appreciated that some or all of the features disclosed in FIGS. **1-8** may be incorporated in the retail display security system **1000**.

(76) The retail display security system **1000** can further include a digital key gateway **416**. The digital key gateway **416** can be connected to all digital key dispensers **404** in a store using a long range, private wireless network **1010** as illustrated in FIG. **4B**. It can also optionally be connected to the internet. The digital key gateway **416** enhances the system security.

(77) First, the digital key gateway **416** can include serve information to a monitoring console associated with graphical user interface **1025** that is accessible by store associates and can provide information. The information can include a door sensor **1030** signal indicating a fixture door has been left open with an open door alert **1105**. Camera or video data from camera **1020** can be used for this purpose as well. The door alert can also include a timer to collect information such as the amount of time the fixture door remained open each time it was unlocked.

(78) The digital key gateway **416** can further monitor stocking levels in the product display **645** with a scale **1015** integrated in the display and/or via camera **1020**. When the stocking levels become low, an inventory monitor **1110** can send an alert that the display needs re-stocking. The digital key gateway **416** can also monitor battery life, and can notify a store admin when a new battery is required.

(79) The digital key gateway **416** can also be used to detect potential nefarious shopper behavior such as removal of numerous products when the door is open, with a theft alert module **1115**. For example, the scale **1015** can be calibrated with the weight of a given product. If more than one product is removed while the door is open, the associated weight will be incorrect, and a notification can be provided to the admin. In certain embodiments, where a shopper may wish to purchase two of a product, a prompt can be provided in the user portal requiring the user to indicate the number of product they will remove.

(80) Likewise, location data, sensor data, and temporal data collected from a user device enable the AI module **915** to detect potential nefarious usage patterns associated with activity within a specific radius from a given location.

(81) Various advantages can be realized using the systems and methods disclosed herein. As noted, users may not need to install an app on their mobile device. The embodiments takes advantage of native mobile device apps such as a web browser. The digital key dispenser can communicate with the smart lock wirelessly. Aspects of the embodiments, including the digital key dispenser and the door lock can be powered by batteries to reduce installation cost, complexity, and maintenance. No hard wiring required, although in certain embodiments, some or all of the systems can be connected to AC power.

(82) Likewise, the digital key dispenser and the wireless door lock do not require an internet connection of any kind. The access module **630** is accessed using the shopper's mobile device internet connection.

(83) FIG. **13** illustrates steps associated with a method **1300** for theft deterrence in accordance with the disclosed embodiments. The method begins at step **1305**.

(84) At step **1310** products can be displayed in a retail store display which can be locked with a smart lock. When a shopper wishes to purchase an item in the display, at step **1315**, the shopper can scan or otherwise enter a code provided on a digital key dispenser with an internet connected device. At step **1320** the code directs the device to a web page where the shopper can enter identity confirming information such as the shopper's name, phone number, address, and/or email address.

The shopper may also be prompted to select the door they wish to unlock.

(85) At step **1325**, the shopper information can be stored. Next at step **1330** a single use digital key can be provided to the shopper on the device. The shopper can scan the single use key with a digital key reader associated with the digital key dispenser at step **1335**. If the key is a valid key, the smart lock associated with the retail store display can be disengaged at step **1340** so that the shopper can remove the desired product at step **1345**. The method ends **1350**.

(86) FIG. **14** illustrates another embodiment of a product access system **1400** in accordance with the disclosed embodiments. Aspects of the product access system may include some or all of the features disclosed in other embodiments illustrated herein. The product access system **1400** generally includes a display **1402** which can comprise a product display and/or a retail product display. In certain embodiments, the display **1402** can include one or more doors fitted with locks, which provide access to products therein. In FIG. **14** four doors are illustrated first door **1404**, second door **1406**, third door **1408**, and fourth door **1410**, but in other embodiments a different number of doors can be used. The doors are locked, so that the products stored in the display are not readily accessible.

(87) The display **1402** can include a dispenser **1412**, configured to allow a shopper **1414** to access products in the display **1402**. The dispenser **1412** can comprise a printed code **1416** displayed on the dispenser or a digitally rendered code displayed in a display window on the dispenser. In certain embodiments, the printed code **1416** can comprise a QR code, bar code, string of characters, or other such code. The dispenser **1412** further includes a keypad **1418**.

(88) In order to access the products in the display **1402**, the shopper can scan the printed code **1416**, for example, by using a camera on their phone to capture the code. Scanning the printed code **1416** automatically launches a Mobile UI **1420** on the mobile device the shopper **1414** used to scan the code. From the mobile UI **1420** the shopper **1414** can specify the door on the display they want to open. The shopper **1414** is then prompted via the mobile UI **1420** displayed on the shopper's phone to enter the shopper's phone number. Once the shopper enters their phone number, the code generator **1422** in the access cloud **1428** can provide a code to the shopper **1414** via the mobile UI **1420** and/or as a "text message" via the SMS protocol at the phone number entered by the shopper **1414**. The shopper **1414** can then enter the code using the keypad **1418** on the dispenser **1412**. In certain embodiments, the code comprises a digital key code (e.g., a string of numbers). If the shopper **1414** enters the correct code and the code is authentic, one of the doors will unlock, allowing the shopper **1414** to remove product from the display **1402**. If the door is left open an alarm can be triggered to alert personnel that the door needs to be closed and locked. If the door is not left open the lock on the door is reengaged automatically. In certain embodiments, the lock can include an optical sensor as further detailed herein.

(89) The system can include an access cloud **1428**, which can include a web portal **1430** that serves as the access point for non-shopper actors associated with the system. The web portal **1430** is operably connected to a user authentication module **1432**, which can be further connected to a storage structure or database **1434**, as well as the code generator **1422** and mobile UI **1420**. This architecture is exemplary and can be organized in other ways in other embodiments.

(90) The disclosed embodiments can be used by a retailer **1426** and one or more specific stores **1436** operated by the retailer **1426**. The retailer **1426** can assign an administrator **1438**. Any one of the stores **1436** is managed by a store manager **1446** and may be operated by one or more store associates **1440**. The administrator **1438** and the store manager **1446** can access the web portal **1430** via desktop computers, laptop computers, tablet computers, mobile devices or the like. The store associates **1440** can access the web portal **1430** by an associate application **1442** available on an associate mobile device **1444**.

(91) Administrators **1438** can be given full read-write access to all the stores **1436** in the retail chain **1426**. The administrator **1438** can control manager **1446** and associate **1440** access to the access cloud **1428** and can perform any tasks those users can.

(92) Managers **1446** can control associate **1440** access to the access cloud **1428** and can perform any tasks an associate **1440** can. Managers **1446** can monitor the health of all the fixtures (e.g., displays **1402**) in their store(s) **1436**, track who restocked each fixture and when, and track shopper activity.

(93) The system **1400** is designed to dramatically reduce or eliminate associate **1440** assistance for shoppers **1414** to access products secured in the display **1402**. However, shelves in the display **1402** still need to be re-stocked. Similarly, shoppers **1414** without a cell phone, or those uncomfortable with technology, may require assistance to unlock a display **1402**. Associates **1440** perform these tasks using the associate app **1442**. Associates **1440** may also be given limited access to the web portal **1430**.

(94) Aspects of the system **1400** can be monitored by a system administrator **1448**. The system administrator **1448** can work for an independent third party. They are typically members of a compliance team. The system administrator **1448** can access the web portal **1430** and, after authentication with the authentication module **1432**, will have full read-write access to the entire access cloud **1428**, and control user access to the system **1400**.

(95) The third party may also provide one or more provisioning technicians **1450**. The provisioning technicians **1450** can use a provisioning application **1452** on the provisioning technician's mobile device **1454** to access the web portal **1430** and the dispenser **1412**. The provisioning application **1452** allows the provisioning technician **1450** to provision the dispenser **1412**. The provisioning application **1452** can pair dispensers **1412** with doors on the display **1402**. The provisioning application **1452** can also be used for troubleshooting, and testing dispenser **1412** hardware. This provisioning application **1452** can be designed to authenticate the provisioning technician **1450** with the user authentication module **1432** without being part of a retailer organization **1426**.

(96) The provisioning application **1452** can be configured to be operated by third party personnel, such as the provisioning technician **1450** or the system administrator **1448**. The provisioning application **1452** allows users to connect to dispensers **1412** wirelessly, and perform various maintenance and configuration tasks.

(97) Users that open the provisioning application **1452** can initially be presented with a login screen. All users are authenticated by the Cloud **1428** with the user authentication module **1432**. Users can reset their password from the login screen. This can be initiated by the Cloud **1428** sending an email to the user with a reset link.

(98) The following description illustrates exemplary uses of the provisioning application **1452**. After authentication, a user of the provisioning application **1452** can pair the app **1452** with a dispenser **1412** by entering pairing mode on the dispenser **1412**, scanning for dispenser BLE advertisements, and selecting the desired dispenser **1412** (realizing there may be multiple dispensers dispersed in a store **1436**). The list of dispensers can be sorted in descending received signal strength indicator "RSSI" order (i.e., strongest RSSI at the top).

(99) The dispenser **1412** enters pairing mode for 30 seconds after power is applied. While in pairing mode, the MAC address is displayed on the dispenser screen as a QR code. In other embodiments, the dispenser **1412** can enter pairing mode in other ways. For example, while holding the "CLR" key down on the keypad **1418**, the user can press and hold the "1" key. An LED will blink red (e.g., for 5 seconds). Next, the "1" key is released followed by the "CLR" key.

(100) In certain embodiments, all dispensers **1412** in a store **1436** can advertise their BLE name in the following format: "DISP-XXXXXXXXXXXX". where the 12 "Xs" represent the BLE MAC address of the dispenser (without colons separating the 6 octets).

(101) The user can then bind/link doors with dispensers **1412**. Two methods can be used for this step. The first is by scanning the BLE MAC address barcode/QR code label on the door lock with a mobile phone camera. The other is by selecting the door lock from a sorted lock list constructed by scanning for advertisements. The list can be sorted in descending RSSI order (i.e., strongest RSSI at the top). It should be appreciated that, in certain embodiments, all door locks continuously

advertise their name when not connected to dispenser **1412**. They can use a name format is as follows: “LOCK-XXXXXXXXXXXX”, where the 12 “Xs” represent the BLE MAC address of the lock (without colons separating the 6 octets).

(102) The provisioning application **1452** can also provide identifying characteristics for all dispensers **1412** and associated doors in the store **1436**. This information can include BLE MAC address, device name, battery level percentage, firmware revision, hardware revision, retailer ID, store ID, and the like.

(103) Using the provisioning application **1452**, users can also read and write (if writable) all of the characteristics defined in the BLE custom service for both the dispenser **1412** and associated door (and locks). Once a dispenser **1412** is successfully paired, the user can save the dispenser configuration in the provisioning application **1452** for reuse. Dispenser configurations saved in the app can be downloadable to other dispensers **1412**.

(104) The provisioning application can also be used for testing and maintenance. For example, users can open any door associated with any dispenser **1412** that has been instantiated in the provisioning application **1452**. In this case, the application **1452** can communicate with the door (or lock) directly. Users can open all doors associated with a specific dispenser **1412** with a single tap.

(105) Likewise, users can update the dispenser **1412** and/or door (lock) firmware over-the-air. The provision application can facilitate this function with access a cloud storage location containing firmware files that allow the user to select a specific file.

(106) The following description illustrates exemplary uses of the associate application **1442**. The associate application **1442** is primarily intended for use by managers **1446** and associates **1440**. However, administrators **1438** may also be trusted users of the associate application **1442**, provided they have an active account.

(107) There are three primary uses functions for the associate application **1442**. Before any such functions can be realized or actions can be performed, associate application **1442** users must first be authenticated. Users of the associate application **1442** are first presented with a login screen. All users are authenticated by the cloud **1428** and user authentication module **1432** before the associate application **1442** can be accessed. This ensures the user is only granted access to the objects appropriate for their role(s).

(108) Associate application **1442** users can be assigned one or more roles. Users can reset their password from the login screen. This is achieved by the Cloud sending an email to the user with a reset link.

(109) At the associate application **1442** home screen a user is able to see all the displays **1402** they have been granted access to in the store **1436**. The store number, display name, and status of each display **1402** can be made immediately visible. The user can filter and sort by display names. The display **1402** status can be maintained by the cloud **1428** and, in an exemplary embodiment can have three states: 1) Ok (green); 2) Service Soon (yellow); and 3) Immediate Attention (red).

(110) In addition, associate application **1442** can display the app version and username. The user can access their Cloud profile to change their password as necessary and, if the user has been given permission to perform one or more maintenance tasks, they are able to access the maintenance screen(s) from the home screen of the associate application **1442**.

(111) In an exemplary embodiment, the associate application **1442** can be used by the associate **1440** to provide shopper assistance. For this use case, the users can open doors on the display **1402** for shoppers with no more than 2 taps (after they have logged in). The user can scan the QR code **1416** on the dispenser **1412** E-Ink screen, from within the app, and will immediately be able to select a door to open.

(112) In another embodiment, the associate application **1442** can be used for restocking. While similar to the shopper assistance use case, it is different because when a door is opened for restocking the “door left open” alarm is disabled. When restocking is complete, and the door is

closed it automatically re-enables the “door left open” alarm.

(113) In another embodiment, the associate application **1442** can be used for maintenance. In this case, the user can add displays **1402** to the stores **1436**. Users may access only the stores **1436** their account is associated with. Users can similarly add a display **1402** to a store **1436** by giving it a descriptive name. The user is able to bind/link dispensers **1412** to displays **1402**. This is accomplished by scanning the MAC address displayed on the screen while in pairing mode. The user can also bind doors to dispensers **1412** by scanning the MAC address sticker on the bottom of the door lock.

(114) In yet another embodiment, the associate application **1442** can be used for troubleshooting problems. The associate application **1442** can provide information related to a display **1402** such as battery level of individual dispensers **1412** and doors. The battery level can be indicated with indicators such as: 1) Ok (>80%); 2) Replace soon (>25% and <80%); 3) Replace immediately (<25%). The associate application **1442** can also be used for configuring low level settings and updating firmware.

(115) The web portal **1430** can be embodied as a hardware or software module, or as a web application as detailed herein. Likewise, the access cloud **1428** can be understood to be a computer architecture with remote or “cloud” storage, used to implement hardware or software applications as detailed herein. The primary responsibility of the access cloud **1428** is to generate and deliver unlock codes to shoppers **1414** via the code generator **1422**. The cloud **1428** can host a web site referred to as the web portal **1430**. Once logged into the web portal **1430**, users can perform tasks based on varying levels of permission.

(116) FIG. **15** illustrates an exemplary screenshot of a home screen **1500** associated with the web portal **1430**.

(117) FIG. **16** illustrates aspects of the dispenser **1412**, in accordance with the disclosed embodiments. The dispenser **1412** includes a housing **1605**, configured to hold a processing PCB and battery not shown. The dispenser is battery powered so that it can operate without WIFI or power in areas of a store **1436** where power and/or WIFI are not readily available (sometimes referred to as a “dark aisle”). The system **1400** is configured to operate at very low power consumption so that it can operate long term without new batteries.

(118) The electronics are operably connected to a display **1620** for displaying a code **1416**. In certain embodiments, the display can be an electronic display that generates a unique code **1416** after the dispenser **1412** is commissioned. In other embodiments, the code can be generated externally and physically affixed to the dispenser **1412** (e.g., as a sticker). A prompt **1615** can be provided adjacent to the display **1620** prompting users to scan the code **1416** in order to open the display. The housing can further include an LED indicator light **1610**, which can light up in multiple colors and/or in various patterns to signify operational status. The dispenser **1412** further includes a keypad **1418** for code entry.

(119) An important aspect of the systems (i.e., system **1400**) disclosed herein, is that the shopper **1414** can remain anonymous. The system **1400** can be configured to collect the shopper **1414** identification only as a phone number which is then hashed, or as a mac address. Thus, while the system serves to deter shoplifting, it also protects the shopper's **1414** privacy since shopper data need not be collected. In certain embodiments, these safeguards can meet Protected Personal Information (PPI) and General Data Protection Regulation (GDPR) requirements.

(120) Another important advantage of the disclosed systems is that they can be easily integrated into existing third party devices or applications. For example, the dispenser can be commissioned on or near any display system which can be locked. The dispenser can be operably connected to the locking mechanism via blue tooth NFC or other such technology and then used, along with the associated software and hardware systems to control access to the third party display.

(121) Another aspect of the disclosed systems can include locking one or more doors of the display **1402** with a lock that uses optical sensor detection. FIG. **17** illustrates aspects of an exemplary

locking system **1700**. The locking system **1700** generally comprises a housing **1705** with a locking slot **1710**. A rotating locking pawl **1715**. The actuator (not shown) for the locking pawl **1715**, can be operably connected to an optical sensor **1720** configured in an aperture **1725** in the housing **1705**. In operation, the optical sensor **1720** is configured to actuate the locking pawl **1715** when an optical input is interrupted (e.g., when the door to the display blocks the path of an optical signal to the optical sensor. In certain embodiments, the optical sensor can therefore be aligned in the optical path of, for example, a low power laser or other such optic. The optical path is arranged such that the door interrupts the optical path when it is closed. When the optical sensor is no longer receiving the optical input the lock **1700** will lock. When the door is not interfering with the optical path, the lock will remain unlocked.

(122) It should be understood that, in other embodiments, the lock **1700** can work in the opposite way. In other words, the door can unlock when the optical sensor stops receiving an input, and can lock when the optical input is once again incident on the sensor.

(123) Based on the foregoing, it can be appreciated that a number of embodiments, preferred and alternative, are disclosed herein. For example, in an embodiment, a display system comprises a display secured with a lock, a dispenser, the dispenser comprising: a digital key request code and an input configured to accept entry of a digital key, and an access cloud configured to transmit the digital key to a user device when the digital key is requested using the digital key request code, wherein the lock is unlocked when the digital key, entered with the input, is authentic. In an embodiment, the lock comprises an electronic lock.

(124) In an embodiment, the digital key request code comprises a QR code. In an embodiment, the digital key comprises a string of numbers. In an embodiment, the digital key comprises a unique digital key every time the digital key is requested using the digital key request code. In an embodiment, the display comprises a retail product display.

(125) In an embodiment, the access cloud further comprises a mobile UI configured to require the user to input their phone number in order to receive the digital key. In an embodiment, the mobile UI provides the digital key to the user after the user enters their phone number. In an embodiment, the access cloud provides the digital key to the user as a text message using the phone number provided by the user.

(126) In another embodiment, a secure display system comprises a display secured with a lock, a digital key request code, an access module configured to transmit a digital key to a user device when a digital key is requested using the digital key request code, and a digital key reader configured to unlock the lock when the digital key is authentic. In an embodiment, the lock comprises an electronic lock. In an embodiment, the display comprises a retail product display.

(127) In an embodiment, the digital key request code comprises a QR code. In an embodiment, the digital key request code is provided proximate to the digital key reader. In an embodiment, the access module generates a unique digital key every time the digital key is requested using the digital key request code. In an embodiment, the digital key comprises a QR code. In an embodiment, the digital key reader comprises a camera configured to scan the QR code.

(128) In another embodiment, a secure access method comprises providing a digital key request code on a dispenser, upon receiving a digital key request, requiring entry of a phone number via a web portal, providing a digital key to a user device, accepting input at the dispenser, the input comprising the digital key, authenticating the digital key, and unlocking a lock associated with a display when the digital key is authentic. In an embodiment, providing a digital key to a user device further comprises at least one of: providing the digital key to the user via a mobile UI and/or providing the digital key to the user as a text message using the phone number entered via the web portal. In an embodiment, the digital key comprises at least one of a QR code and/or a string of numbers.

(129) It should be appreciated that variations of the above-disclosed and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or

applications. It should be understood that various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

Claims

1. A display system comprising: a display secured with a lock; a dispenser, the dispenser comprising: a digital key request code provided on the dispenser and uniquely associated with the display; and an input configured to accept entry of a unique digital key, the unique digital key comprising a code that can be authenticated by an authentication module; and an access cloud configured to transmit the unique digital key to a user device when the unique digital key is requested using the digital key request code, the access cloud further comprising a mobile UI configured to require a user to input a phone number in order to request the unique digital key, wherein the lock is unlocked when the unique digital key, entered with the input on the dispenser, is authentic.
2. The display system of claim 1 wherein the lock comprises an electronic lock.
3. The display system of claim 1 where the digital key request code provided on the dispenser and uniquely associated with the display comprises a QR code.
4. The display system of claim 1 wherein the unique digital key comprises a string of numbers generated by a digital key module, when the unique digital key is requested using the digital key request code.
5. The display system of claim 4 wherein the unique digital key comprises a different unique digital key generated by the digital key module every time the digital key is requested using the digital key request code.
6. The display system of claim 1 wherein the display comprises a retail product display.
7. The display system of claim 1 wherein the digital key request code provided on the dispenser further comprises: a printed code physically attached to the dispenser.
8. The display system of claim 7 wherein, the mobile UI provides the unique digital key to the user after the user enters their phone number using the mobile UI.
9. The display system of claim 7 wherein the access cloud provides the unique digital key, generated by a digital key module, to the user as a text message using the phone number provided by the user via the mobile UI.
10. A secure display system comprising: a digital key request code uniquely associated with a product display having a lock; an access module configured to transmit a digital key, generated by a digital key module, for the lock associated with the product display, to a user device when a digital key is requested using the digital key request code uniquely associated with the product display; and a digital key reader configured to unlock the lock when the digital key is authentic.
11. The secure display system of claim 10 wherein the lock comprises an electronic lock.
12. The secure display system of claim 10 where the digital key request code uniquely associated with a product display comprises a QR code.
13. The secure display system of claim 10 where the digital key request code uniquely associated with a product display is provided proximate to the digital key reader.
14. The secure display system of claim 10 wherein the digital key module generates a unique digital key every time the digital key is requested using the digital key request code uniquely associated with a product display.
15. The secure display system of claim 10 wherein the display comprises a retail product display.
16. The secure display system of claim 10 wherein the digital key, generated by the digital key module, uniquely associated with a product display comprises a printed QR code affixed to the housing of the digital key reader.
17. The secure display system of claim 16 wherein the digital key reader comprises a camera

configured to scan the QR code.

18. A secure access method comprising: providing a digital key request code on a dispenser; in response to receiving a request for a digital key using the digital key request code, generating a digital key with a digital key module, wherein the digital key is uniquely associated with the dispenser; upon receiving a digital key request, requiring entry of a phone number via a mobile UI associated with a web portal; providing the digital key, generated by the digital key module, to a user device, upon receiving the phone number via the web portal; accepting input at the dispenser, the input comprising the digital key; authenticating the digital key; and unlocking a lock associated with a display when the digital key is authentic.

19. The secure access method of claim 18 wherein providing a digital key to a user device further comprises at least one of: providing the digital key to a user via the mobile UI; and/or providing the digital key to the user as a text message using the phone number entered via the web portal.

20. The secure access method of claim 18 wherein the digital key uniquely associated with the dispenser comprises at least one of: a QR code; and/or a string of numbers.
